

	$mass = 5.02 \times 10^{-26} \text{ kg}$
8(a)	Internal energy is the sum of the microscopic Kinetic Energy and Microscopic Potential energy of the molecules.
8(a)(ii)	+q – Thermal Energy supplied to system. +w – Work done on the system
8(b)(i)	Ideal gas is a gas that obeys the equation of state $pV=nRT$, where p = pressure of gas in Pa V = volume of gas in m^3 T = thermodynamic temperature of gas in K n = amount of gas in moles (Wrong to write : number of moles) R = molar gas constant
8(b)(ii)1	Using the ideal gas equation : $pV = nRT$ $pV = NkT$ $N = \frac{pV}{kT} = \frac{5.4 \times 10^5 \times 1.2 \times 10^{-2}}{1.38 \times 10^{-23} (57 + 273.15)} = 1.42 \times 10^{24}$
8(b)(ii)2.	Mean KE of 1 molecule = $\frac{3}{2} kT$ $= \frac{3}{2} (1.38 \times 10^{-23}) (57 + 273.15)$ $= 6.83 \times 10^{-21} \text{ J}$
8(b)(ii)3.	The total internal energy consist of kinetic energy only as there is no potential energy of the molecules. Total internal Energy: $\frac{3}{2} NkT$ $= \frac{3}{2} (1.38 \times 10^{-23}) (57 + 273.15) (1.42 \times 10^{24})$ $= 9704 \text{ J}$

8(c)(i)	Total initial internal Energy: $\frac{3}{2}NkT$ $= \frac{3}{2}(1.38 \times 10^{-23})(57 + 273.15)(1.42 \times 10^{24})$ $= 9704\text{J}$ Total final internal Energy: $\frac{3}{2}NkT$ $= \frac{3}{2}(1.38 \times 10^{-23})(155 + 273.15)(1.42 \times 10^{24})$ $= 12585\text{J}$ Total increase in internal energy : 2880 J <u>Shorter method:</u> Increase in internal energy $\Delta U = \frac{3}{2}Nk\Delta T$ $= \frac{3}{2}(1.42 \times 10^{24})(1.38 \times 10^{-23})(155 - 57)$ $= 2880\text{J}$
8(c)(ii)1.	The volume has to be kept constant such that work done on system is zero. According to first law of Thermodynamics, increase in internal energy would then be equal to thermal energy supplied.
8(c)(ii)2.	Since total increase in internal energy is equivalent to the heat supplied. $Q = nc\Delta\theta$ $n = \frac{\text{no of molecules}}{\text{avogadro}} = \frac{1.42 \times 10^{24}}{6.02 \times 10^{23}} = 2.36 \text{ mol}$ $c = \frac{Q}{n\Delta\theta} = \frac{2881}{2.36(155 - 57)} = 12.46 \text{ J mol}^{-1} \text{ K}^{-1}$
9(a)(i)	Simple harmonic motion is defined as an oscillatory motion in which the acceleration